

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR.

CS XI

Chapter 1: Computer Systems

Lecture 10 of 20

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

LEARN IT

A logic expression (Boolean expression) combines variables with operations, for example $Y = A + B' \cdot C$. To evaluate it correctly you must follow the **order of precedence**, just like BODMAS in maths:

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

WORKED EXAMPLE

Worked example 1: truth table of $Y = A + B' \cdot C$

Step 1: 3 inputs, so 8 rows.

Step 2: Precedence: first find B' (NOT), then $B' \cdot C$ (AND), then $A + (B' \cdot C)$ (OR). Make a column for each step.

A	B	C	B'	$B' \cdot C$	$Y = A + B' \cdot C$
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

Step 3: $Y = 1$ in rows 1, 4, 5, 6, 7 (decimal). Check one row by hand: $A=0, B=0, C=1 \rightarrow B' = 1, B' \cdot C = 1, Y = 0 + 1 = 1$. ✓

WORKED EXAMPLE

Worked example 2: truth table of $Y = (A + B) \cdot C'$

Step 1: Parentheses first: find $A + B$. Then NOT: C' . Then AND the two.

A	B	C	$A+B$	C'	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Step 2: $Y = 1$ in rows 2, 4 and 6 (decimal).

Minterms (SOP) and Maxterms (POS)

LEARN IT

Sometimes you are given a truth table and must *find* the expression. There are two standard ways.

Minterm (used in SOP — Sum of Products): a product (AND) of all the variables, written for a row whose output is 1. A variable is **complemented (A')** if its value is 0, and **uncomplemented (A)** if its value is 1. OR-ing all the minterms gives the SOP expression.

Maxterm (used in POS — Product of Sums): a sum (OR) of all the variables, written for a row whose output is 0. A variable is **complemented (A')** if its value is 1, and **uncomplemented (A)** if its value is 0. AND-ing all the maxterms gives the POS expression.

A	B	Output (Y)	Minterm (SOP)	Maxterm (POS)
0	0	0	–	$A + B$
0	1	1	$A'B$	–
1	0	0	–	$A' + B$
1	1	1	AB	–

Final SOP expression: $Y = A'B + AB$. Final POS expression: $Y = (A + B)(A' + B)$. Both describe the same table. (Both simplify to $Y = B$ — look at the table: Y always equals B. You will confirm this with a K-map later.)

WORKED EXAMPLE

Worked example 3: write the SOP for a 3-input "majority" circuit. $Y = 1$ when at least two of A, B, C are 1.

Row	A	B	C	Y	Minterm
0	0	0	0	0	–
1	0	0	1	0	–
2	0	1	0	0	–
3	0	1	1	1	$A'BC$
4	1	0	0	0	–
5	1	0	1	1	$AB'C$
6	1	1	0	1	ABC'
7	1	1	1	1	ABC

Step 1: Pick the rows with $Y = 1$: rows 3, 5, 6, 7.

Step 2: Row 3 is $A=0, B=1, C=1 \rightarrow A$ is 0 so write A' ; B and C are 1 so write B, C $\rightarrow A'BC$.

Step 3: Do the same for the other rows and OR them: $Y = A'BC + AB'C + ABC' + ABC$. In short form: $Y = \Sigma m(3, 5, 6, 7)$. We will simplify this with a K-map in 1.1.13.

$Y = A'BC + AB'C + ABC' + ABC = \Sigma m(3, 5, 6, 7)$

REMEMBER IT

SOP = 1s, POS = 0s. Minterm: 0 $\rightarrow$ bar, 1 $\rightarrow$ plain. Maxterm: the reverse (1 $\rightarrow$ bar, 0 $\rightarrow$ plain). Think "Min likes 1s, Max likes 0s".

COMMON MISTAKE

Two classic errors: (1) writing minterms for the 0-rows — minterms come only from rows where $Y = 1$; (2) forgetting precedence — in $A + B \cdot C$ you must do $B \cdot C$ first. $A + B \cdot C$ is **not** $(A + B) \cdot C$: for $A=1, B=0, C=0$ the first gives 1 and the second gives 0.

اردو میں سمجھیں

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR۔ مینٹرم (Minterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 1 ہو؛ اس میں 0 والے متغیر پر بار (A') لگتی ہے۔ میکس ٹرم (Maxterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 0 ہو؛ اس میں 1 والے متغیر پر بار لگتی ہے۔

CHECK YOURSELF

1. Which comes first: AND or OR?

2. Write the minterm for $A = 1, B = 0, C = 1$.

3. Write the maxterm for $A = 0, B = 1$.

Answers: 1. AND. 2. $AB'C$. 3. $A + B'$.

Next lecture: 1.1.11 Logic Diagrams